Master Research Status: Electric Cable Shovel Digital Twin (M1, M2, and M3)

Project: IIT/SRIC/R/AEH/2026/104 (CMPDI) — *AI-Enabled Diagnosis, Prognosis and Health Management of Mining Electrical Systems* **Research Topic:** *Dynamic Modelling and Digital Twin of Electric Cable Shovels: Physics-Based External-Load Estimation, Payload Estimation and Machine Health Monitoring* **Researcher:** Suhail Majeed Sheikh, Junior Research Fellow, Department of Mining Engineering, IIT Kharagpur **Supervisor:** Prof. Sunita Mishra **Master Status Date:** 2026-11-06

1. Milestone Readiness Scorecard (M1 – M3)

Milestone	Scope & Technical Objective	Primary Scripts & Models	Verified .mat Archives	Current Status	Key Quantitative Benchmark
Milestone M1	2-DOF Euler--Lagrange dipper--handle forward validation against Simscape Multibody & blind 5-parameter OLS identification	shovel_params.m, validate_and_identify.m, ShovelSimulatorr2025b.slx	3 (stage1_results.mat, run_trim05.mat, run_notrim.mat)	**FROZEN & COMPLETE**	Forward R^2 = 1.000000 (1.35\times 10^{-6}\ \mathrm{N\cdot m}, 8.23\times 10^{-11}\ \mathrm{N}); all 5 parameters recovered to 0.000\% error with sensed \ddot{q}; +56.1\% I_{zz} bias under gradient velocity differentiation
Milestone M2	Known external point-force injection Q_{\mathrm{ext}} = J_s(q)^T F, linearity/frequency/configuration sweeps, station confounding, formal SVD	recover_load.m, sensitivity_diagnostic.m, plot_m2.m, m2_formal_identifiability_svd.m, m2_p05_config_sweep.m, m2_p07_model	25 active (P00: 2, P01: 1, P02: 2, P03: 6, P04: 7, P04b: 6, P06: 3)	**ANALYSIS COMPLETE** (Local Simulink .slx runs optional for P05 grid & torque-driven P08)	Bail load recovery R^2 = 1.000000, \mathrm{NRMSE}_{q_3} \le 4.28\times 10^{-10}\%; station discriminating fraction 1-R^2 collapses

Milestone	Scope & Technical Objective	Primary Scripts & Models	Verified .mat Archives	Current Status	Key Quantitative Benchmark
	identifiability, perturbation fingerprints, & held-out validation (P00–P08)	_perturbation.m, m2_p08_held_out_validation.m, M2_LoadInjection.slx			89.8\times on Bi et al. narrow band (\kappa_2(\tilde{S}) = 497.71, \theta_{\min} = 0.2302^\circ), resolved via joint 2-channel (\tau_3, f_4) estimation (\kappa_2 = 3.77, \theta_{\min} = 29.70^\circ)
Milestone M3 (Lane B)	Two-branch 30\ \mathrm{Hz} generalized-momentum external-load observer eliminating joint acceleration \ddot{q} (Branch A: friction-aware RK4 M=67; Branch B: friction-uncertainty exact FOH)	m3_observer_suite.m (v2.0.0-PRELIMINARY), run_m3_experiments_v2.py (v2.0.0-PRELIMINARY)	M3_observer_audit_results.mat + verified across all 20 M2 .mat archives	**PRELIMINARY (Code-to-Report Audited)**	Zero-load \mathrm{RMSE}_3 = 53.47\ \mathrm{N\cdot m} (Br. A) vs. 300.43\ \mathrm{N\cdot m} (Br. B); 100 kN bail sinusoidal tracking \mathrm{NRMSE} = 1.151\% (peak-to-peak) / 2.27\% (peak) with 50.05\ \mathrm{ms} lag; 100% friction-discrepancy bound coverage

2. Complete Index of Generated Deliverables and Documentation

1. **MASTER_RESEARCH_STATUS.md**: Executive status across M1, M2, and M3.
2. **LITERATURE_TO_M2_M3_MAPPING.md** (research_documentation/literature_to_m2_m3_mapping.md): Detailed mapping of key papers from papers_library.bib and papers.md to M1–M3.

3. **M2_CURRENT_STATE_AUDIT.md** (M2/research_documentation/M2_current_state_audit.md): Forensic audit of all 28 .mat files and MS02P00–MS02P08 packages.
4. **M2_EXPERIMENT_REGISTER.xlsx**: Multi-sheet Excel workbook registering all M2 packages, all 28 .mat archives with SHA-256 hashes, and the 16-trajectory SVD conditioning table.
5. **M2_FORMAL_IDENTIFIABILITY_REPORT.md**: Formal sensitivity matrix SVD, unit-norm vs. physically scaled conditioning \kappa_2(S), VIF, and subspace-angle analysis.
6. **M2_P05_CONFIGURATION_SWEEP_REPORT.md**: Two-channel Jacobian observability maps across (q_3, d_4) and quadrant-binned verification.
7. **M2_P06_GEOMETRY_STATION_REPORT.md**: Station reconciliation (r_{\mathrm{COG}} = d_4 - 1.32\ \mathrm{m} vs. s_{\mathrm{bail}} = d_4 + 1.50\ \mathrm{m}) and wrong-station regression verification.
8. **M2_P07_MODEL_PERTURBATION_REPORT.md**: 15-case parameter perturbation fingerprint library and least-squares basis attribution.
9. **M2_P08_HELD_OUT_VALIDATION_REPORT.md**: 20-archive cross-trajectory held-out validation for both M2 algebraic recovery and M3 30\ \mathrm{Hz} momentum observers.
10. **M3_FILE_MIGRATION_REGISTER.xlsx**: Excel workbook tracking all 19 M3 artifacts across M3/00_README through M3/10_ARCHIVE.
11. **M3_CODE_AUDIT_REPORT.md**: Complete code-to-report audit (AUD-01 through AUD-07) and Euler--Lagrange \dot{D} = C + C^T verification.
12. **M3_VALIDATION_PROTOCOL.md**: Formal 7-gate data ingestion check, 8-test verification matrix (Tests A–G), and promotion criteria.
13. **CHANGELOG.md**: Complete chronological record of all M1, M2, and M3 additions and audits.
14. **Four New M2 MATLAB Scripts**: m2_formal_identifiability_svd.m, m2_p05_config_sweep.m, m2_p07_model_perturbation.m, and m2_p08_held_out_validation.m.